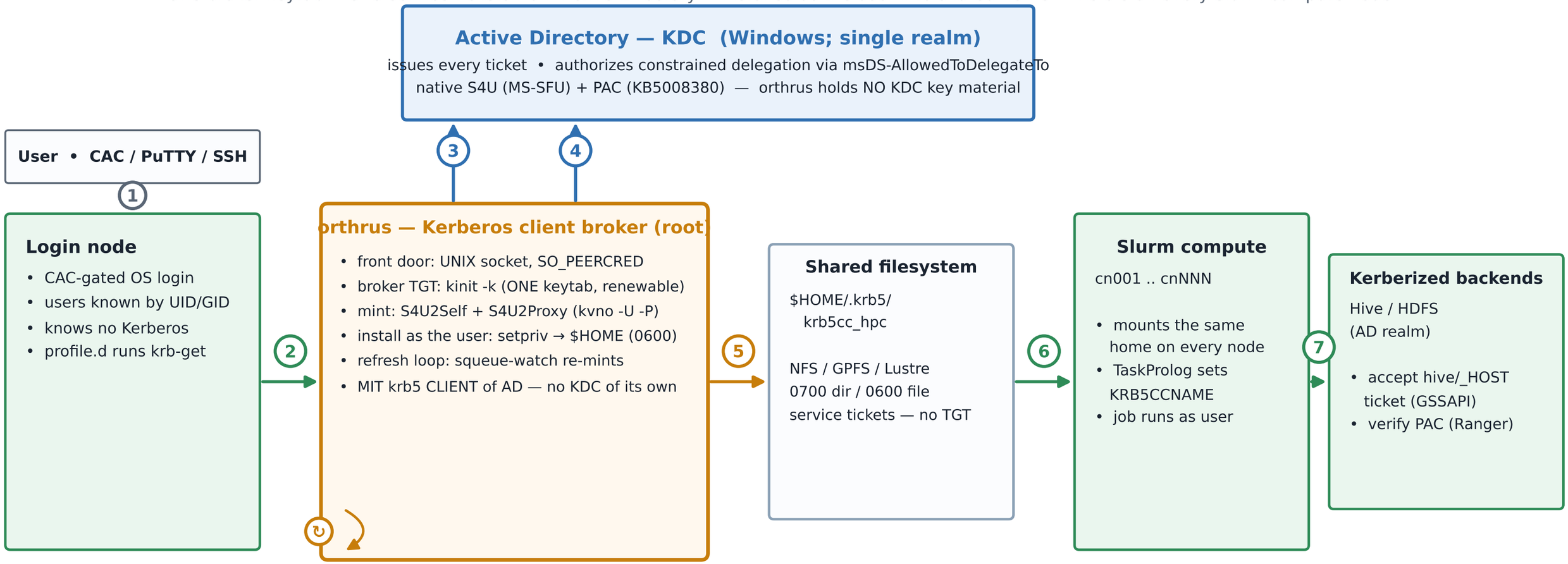
orthrus — Kerberos Client Broker (root): System Architecture

One broker keytab → S4U service tickets from Active Directory → installed as the user into shared-home → visible on every Slurm compute node



End-to-end flow

1. CAC / PuTTY / SSH login establishes the user's UID/GID on the login node (no Kerberos on the HPC side).
2. profile.d runs krb-get → orthrus over a UNIX socket; the UID comes from SO_PEERCRED (kernel-verified, never the request).
3. orthrus keeps its broker TGT fresh: kinit -k from ONE keytab (renewable, unattended) — obtained from AD.
4. orthrus mints the user's SERVICE tickets: S4U2Self + S4U2Proxy (kvno -U user -P spns); AD authorizes via msDS-AllowedToDelegateTo.
5. orthrus installs the cache AS THE USER (setpriv, atomic, 0700/0600) into $HOME/.krb5/krb5cc_hpc on shared storage.
6. Every compute node mounts the same home; the Slurm TaskProlog points KRB5CCNAME at the cache — no copy, no node daemon.
7. The job uses the hive/_HOST (etc.) service ticket over GSSAPI; the backend verifies it and checks the PAC (Ranger).

↻ The squeue-watch refresh loop RE-MINTS service tickets before expiry, so long or queued jobs never find them stale.